Supporting Information: title to be chosen

2026-11-13

Table of contents

Methods 1
- Baselines 1
- Foundation model setup and fine-tuning 2
- Synthetic data generation 2
- Scoring 2
- Implementation 2

Results 3
- Held-out synthetic data 3
- German nowcast hub 3
- Ebola hub submissions 3

References 3

Methods

Baselines

> **Plan**
>
> - `randomwalk` and `randomwalknowcast` in full.
> - Key references: (Johnson et al. 2025).

Foundation model setup and fine-tuning

Plan

- How Chronos-2 and LoRA work, in brief, with schematics.
- Foundation model setup and fine-tuning settings.
- Key references: (Ansari et al. 2025; Hu et al. 2021).

Synthetic data generation

Plan

- Synthetic data generation in full, with an overview of the epinowcast model used.
- Key references: (Abbott et al. 2026; Lison et al. 2024; Park et al. 2024; Abbott 2026).

Scoring

Plan

- Scoring, and its check against scoringutils.
- Key references: (Bosse et al. 2022; Jordan, Krüger, and Lerch 2019; Zanetta and Allen 2024).

Implementation

Plan

- The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
- The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

> **Plan**
>
> - Score breakdowns by location and horizon for each case study.
> - WIS over time for each case study.
> - Held-out synthetic scores by number of fine-tuning outbreaks.

Held-out synthetic data

German nowcast hub

Ebola hub submissions

References

Abbott, Sam. 2026. "Primarycensored: Primary Event Censored Distributions in R and Stan." Zenodo. https://doi.org/10.5281/zenodo.13632838.

Abbott, Sam, Adrian Lison, Sebastian Funk, Carl Pearson, Hugo Gruson, Felix Guenther, Michael DeWitt, James Mba Azam, and Jessalyn Sebastian. 2026. *Epinowcast: A Bayesian Framework for Real-Time Infectious Disease Surveillance*. https://doi.org/10.5281/zenodo.5637165.

Ansari, Abdul Fatir, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, et al. 2025. "Chronos-2: From Univariate to Universal Forecasting." arXiv. https://doi.org/10.48550/arXiv.2510.15821.

Bosse, Nikos I., Hugo Gruson, Anne Cori, Edwin van Leeuwen, Sebastian Funk, and Sam Abbott. 2022. "Evaluating Forecasts with Scoringutils in R." arXiv. https://doi.org/10.48550/arXiv.2205.07090.

Hu, Edward J., Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. "LoRA: Low-Rank Adaptation of Large Language Models." arXiv. https://doi.org/10.48550/arXiv.2106.09685.

Johnson, Kaitlyn E, Maria L Tang, Emily Tyszka, Laura Jones, Barbora Nemcova, Daniel Wolffram, Rosa Ergas, et al. 2025. "Baseline Nowcasting Methods for Handling Delays in Epidemiological Data." *Wellcome Open Research* 10: 614. https://doi.org/10.12688/wellcomeopenres.25027.2.

Jordan, Alexander, Fabian Krüger, and Sebastian Lerch. 2019. "Evaluating Probabilistic Forecasts with **scoringRules**." *Journal of Statistical Software* 90 (12). https://doi.org/10.18637/jss.v090.i12.

Lison, Adrian, Sam Abbott, Jana Huisman, and Tanja Stadler. 2024. "Generative Bayesian Modeling to Nowcast the Effective Reproduction Number from Line List Data with Missing

Symptom Onset Dates." Edited by Tom Britton. *PLOS Computational Biology* 20 (4): e1012021. https://doi.org/10.1371/journal.pcbi.1012021.

Park, Sang Woo, Andrei R. Akhmetzhanov, Kelly Charniga, Anne Cori, Nicholas G. Davies, Jonathan Dushoff, Sebastian Funk, et al. 2024. "Estimating Epidemiological Delay Distributions for Infectious Diseases." medRxiv. https://doi.org/10.1101/2024.01.12.24301247.

Zanetta, Francesco, and Sam Allen. 2024. "Scoringrules: A Python Library for Probabilistic Forecast Evaluation." https://github.com/frazane/scoringrules.